

IT Service Desk (Jev)

Run guide: how to start it, put it online, use it, and test it with real problems

Version	Phase 4 + go-live fixes, 30 Sep 2026
For	Whoever runs or demos the service desk
Stack	FastAPI (API) · Streamlit (web app) · SQLite or Postgres · TypeSafe Jev · RapidOCR
Tests	66 automated tests, all passing

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1. Before you start

You need	Check with	Notes
Windows 10/11 (or Linux/macOS)	–	All commands below are for PowerShell, run from the <code>servicedesk_jev</code> folder
Python 3.12	<code>python --version</code>	Needed only for running on your PC. A server with Docker does not need it
A TypeSafe API key	<code>console.typesafe.ai</code>	This is what lets the bot understand messages. Keep it secret
Docker Desktop (optional)	<code>docker version</code>	Only for the production setup in section 4

The project is two programs that run side by side: the **API** (all the logic and the database, port 8000) and the **web app** (what people see, port 8501). The web app talks only to the API, so start the API first.

2. Run it on your PC

First time only

1. Open PowerShell in the project folder:

```
cd "C:\Users\SANJITH G\OneDrive\Desktop\It\servicedesk_jev"
```

2. Create the Python environment and install the packages (takes a few minutes):

```
python -m venv .venv
.\venv\Scripts\python.exe -m pip install -r requirements.txt
```

3. Create your settings file from the example, then open it and fill it in:

```
Copy-Item .env.example .env
notepad .env
```

Setting	What to put
<code>TYPESAFE_API_KEY</code>	Your key from the TypeSafe console
<code>TYPESAFE_MODEL</code>	<code>jev-latest</code>
<code>DEMO_PASSWORD</code>	Any password of 12+ characters. Every demo account uses it
<code>JEV MOCK=1</code> (optional)	Uncomment to run without the AI (offline, free, less accurate). Good for trying the screens

Every time

Open **two** PowerShell windows in the project folder.

Window 1: the API

Window 2: the web app

```
.\.venv\Scripts\python.exe -m uvicorn api:app  
--port 8000
```

```
.\.venv\Scripts\python.exe -m streamlit run  
app.py
```

Wait for `Application startup complete` . On the very first start it creates the database and all demo accounts.

Your browser opens `http://localhost:8501` . If not, open it yourself.

Demo accounts

Role	Sign in with	Examples
Employee	EMP1001 – EMP1050	EMP1001 Synthetic User 1
IT supervisor	EMP2001	Lakshmi Narayanan: sees every ticket, reports, user admin
IT agents	EMP2002 – EMP2015	EMP2010 Ananya Iyer and EMP2015 Samuel Thomas: VPN (Network Remote Access)

The password for all of them is your `DEMO_PASSWORD` . Demo accounts exist only on a PC run. Production starts with none.

IT Service Desk

Sign in with your employee ID or work email. IT staff use their employee ID too.

Employee ID or work email

EMP1001

Password

Sign in

> How we use your information

> Can't sign in?

Demo accounts: employees EMP1001–EMP1050 · IT supervisor EMP2001 · IT agents EMP2002–EMP2015 (see README). The password is `DEMO_PASSWORD` in `.env`.

The sign-in page. The same form is used by employees and IT staff.

Which agent handles what

Problem	Queue
VPN, remote access	Network Remote Access: EMP2010, EMP2015
Wi-Fi, office network	Network Operations: EMP2009, EMP2015
Passwords, MFA, account locked	Identity Security
Anything urgent, or when no queue fits	Service Desk Duty Manager (supervisor)

The full list is on the supervisor's **User accounts** page, next to each person.

3. Share it online for a demo (temporary link)

This gives you a public `https://...trycloudflare.com` link to your PC. No account, no domain and no router changes are needed.

1. Start the API and web app as in section 2 and leave both windows open.
2. First time only: download the tunnel tool.

```
New-Item -ItemType Directory -Force "$env:LOCALAPPDATA\cloudflared" | Out-Null
Invoke-WebRequest
https://github.com/cloudflare/cloudflared/releases/latest/download/cloudflared-windows-
amd64.exe `
-OutFile "$env:LOCALAPPDATA\cloudflared\cloudflared.exe"
```

3. In a **third** window, open the tunnel:

```
& "$env:LOCALAPPDATA\cloudflared\cloudflared.exe" tunnel --url http://localhost:8501
```

4. Look for the line containing `https://something-random.trycloudflare.com`. That is your link. Share it together with the demo password.

Things to know. The link works only while your PC is on and all three windows are open. It changes every time you restart the tunnel. Anyone with the link and the password can sign in, so share both only with testers. Each chat message uses TypeSafe credit.

4. Run it for real on a server (Docker)

For a permanent address with HTTPS, backups and monitoring. Full details are in `docs/DEPLOY.md`, with the go-live checklist in `docs/GO-LIVE.md`.

1. Rent a Linux server (2 vCPU, 4 GB RAM, 20 GB disk), install Docker, and point a domain name at it (for example `servicedesk.company.com`).
2. Copy the project to the server (for example with `git clone` from your repository), then create the settings file:

```
cp .env.production.example .env.production
chmod 600 .env.production
nano .env.production          # fill in everything under "required"
```

3. Start everything:

```
alias dc='docker compose --env-file .env.production'
dc up -d --build
dc ps                                # api/ui "(healthy)", migrate "Exited (0)"
```

4. Create your own supervisor account. It prints a one-time password, which you change at first sign-in:

```
dc exec api python manage.py create-user EMP0001 "Your Name" you@company.com supervisor --title
"Service Desk Supervisor"
dc exec api python manage.py check          # must end with OK
```

5. Load people from HR: **User accounts** → **Import from HR (CSV)** in the app. The CSV format is in DEPLOY.md.
6. Before real users arrive, go through GO-LIVE.md: approve the employee-friendly articles, set up e-mail/Teams, watchdog alerts and the healthcheck ping, rotate the TypeSafe key, and review the privacy notice.

Runs in production	Job
proxy (Caddy)	HTTPS; the only open ports are 80/443
ui, api	The web app and the logic
worker	Alert timers, sending e-mail/Teams, daily clean-up of expired data
watchdog (healthwatch.py)	Tells IT when the service desk itself is down
db, migrate, backup	Postgres, database updates at start, a daily backup kept 14 days

5. Using it: employee

Get help in the chat

1. Sign in as an employee (for example EMP1001). The **Chat** page opens.
2. Describe the problem in your own words. Or press + to attach a screenshot of the error: the bot reads it on the server and blurs anything that looks like a password.
3. Answer the bot's short questions (often buttons such as your operating system).
4. Follow the steps. They are marked as things *you* do; anything that needs IT is handed over, not left to you.
5. Press **Yes, it's fixed**, **No, still not working**, **I need help with a step** or **Talk to a human**. After two tries that fail, the chat goes to the right IT team with everything already tried, so you don't have to explain it again.
6. Rate answers with 👍 / 👎 . This tells IT which articles need improving.

The screenshot shows a chat interface with a sidebar on the left and a main chat area on the right. The sidebar has sections for 'Get help' (Chat, Ticket form, My tickets), 'Account' (My account), and 'Synthetic User 1' (EMP1001 - Synthetic Role, Sign out). The main chat area shows a conversation where the user says 'No, still not working'. The bot responds with a message and a list of steps to set up the VPN connection again from scratch. The bot also provides a reason why this helps and asks if the user did that. If the problem persists, the bot offers to bring in a person. The chat is then escalated to a human agent, Ananya Iyer from Network Remote Access, who joins the chat. The chat input field at the bottom has a placeholder text: '+ Describe your issue, or attach a screenshot. Never share passwords or codes.'

A VPN chat after two tries: escalated to Network Remote Access, with a friendly time estimate, and the agent joins the chat.

Other pages

Page	Use it to
Ticket form	Raise a ticket by filling in fields instead of chatting
My tickets	See status and replies, reopen a ticket, and rate it 1–5 once it's resolved
My account	Change your password; Prepare my data downloads everything held about you

Get help

Chat

Ticket form

My tickets

Account

My account

Synthetic User 1

EMP1001 - Synthetic Role

Sign out

My tickets

Show closed and cancelled

P3

TKT-3002

With Network Remote Access team

My VPN won't connect

Open

P3

TKT-3001

With Network Remote Access team

My VPN won't connect

Open

P4

TKT-3003

Ananya Iyer (Network Remote Access)

is working on it

Text from the attached screenshot:
ConnectionChecker
Testyourinternetconnection'scapabilitieswith
theVPN Track

Open

TKT-3002

With Network Remote Access team

Category	Team	Assigned to
VPN & Remote Access	Network Remote Access	—
Opened	Target fix by	
27 Sep 00:28 via Chat	30 Sep 15:00	

Short description: My VPN won't connect

Cancel ticket

Conversation

My VPN won't connect

27 Sep 00:28

Quick question: Exact error shown by VPN client (e.g. Paste exact error text)

27 Sep 00:28

A common VPN error is Error 691

27 Sep 00:29

My tickets.

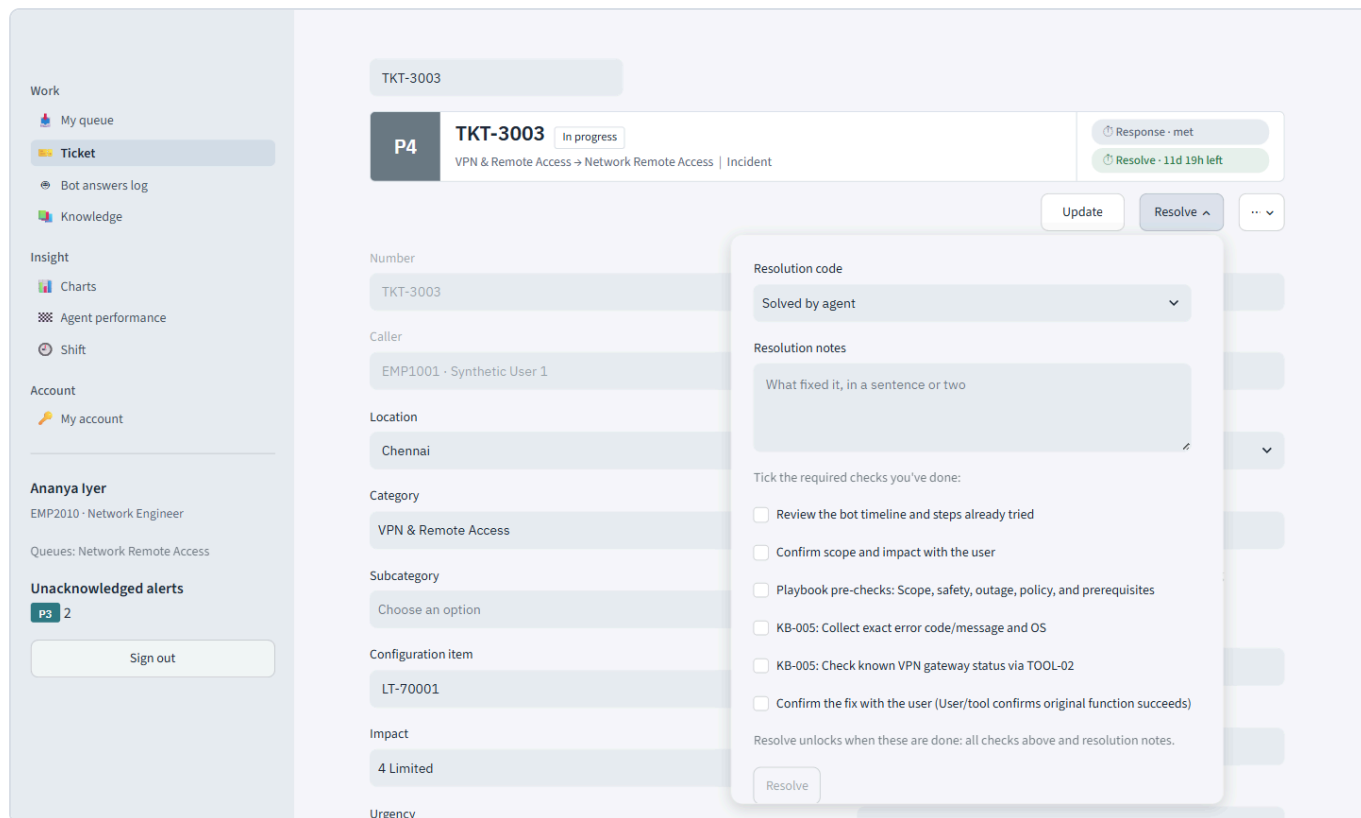
6. Using it: IT agent

1. Sign in as an agent. For VPN tickets that's **EMP2010** (Ananya Iyer).
2. **My queue** lists the tickets in your team's queues. Use **Unowned** to find new work, then click a row to open it.
3. On the ticket, **take it over**. It becomes *In progress* and the employee sees that you've joined their chat.
4. Read the **Bot panel**: what the bot understood, how confident it was, the article it used and why it escalated. **Screenshots** shows the image and the text the bot read from it.
5. Reply to the employee (they see it in the app) or add an internal work note (staff only).
6. Work through the **Checklist**. You can tick items there or in the **Resolve** popup.
7. **Resolve**: choose a resolution code and write one or two sentences on what fixed it. The button unlocks when the required checks and the notes are done.
8. If the bot picked the wrong category, log a **correction**. Corrections are how it gets better.

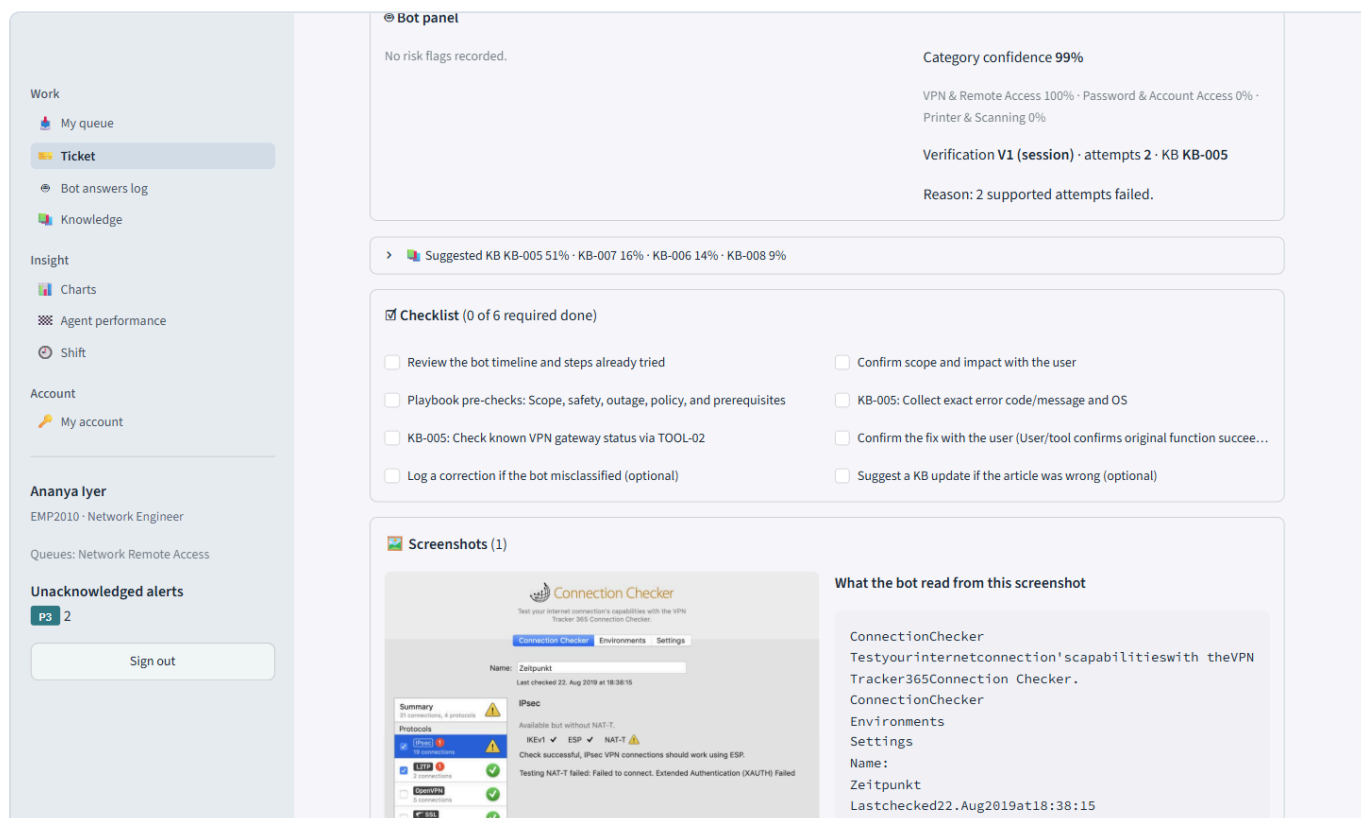
The screenshot shows the 'My queue' interface for an agent named Ananya Iyer (EMP2010 - Network Engineer). The left sidebar contains navigation links for Work (My queue, Ticket, Bot answers log, Knowledge), Insight (Charts, Agent performance, Shift), and Account (My account). The main area displays a table of tickets with columns for Ticket, P, State, Short description, and Queue. The table shows three tickets: TKT-3001 (P3, Escalation queued), TKT-3002 (P3, Escalation queued), and TKT-3003 (P4, In progress). The 'Unowned' filter is selected in the 'Show' dropdown. A 'Log a phone call' button is visible in the top right.

	Ticket	P	State	Short description	Queue
☐	TKT-3001	P3	Escalation queued	My VPN won't connect	Network Remote Access
☐	TKT-3002	P3	Escalation queued	My VPN won't connect	Network Remote Access
☐	TKT-3003	P4	In progress	Text from the attached screenshot: ConnectionChecker Testyourinternetconnection's	Network Remote Access

My queue for a Network Remote Access agent.



The Resolve popup: tick the required checks, add notes, then resolve.

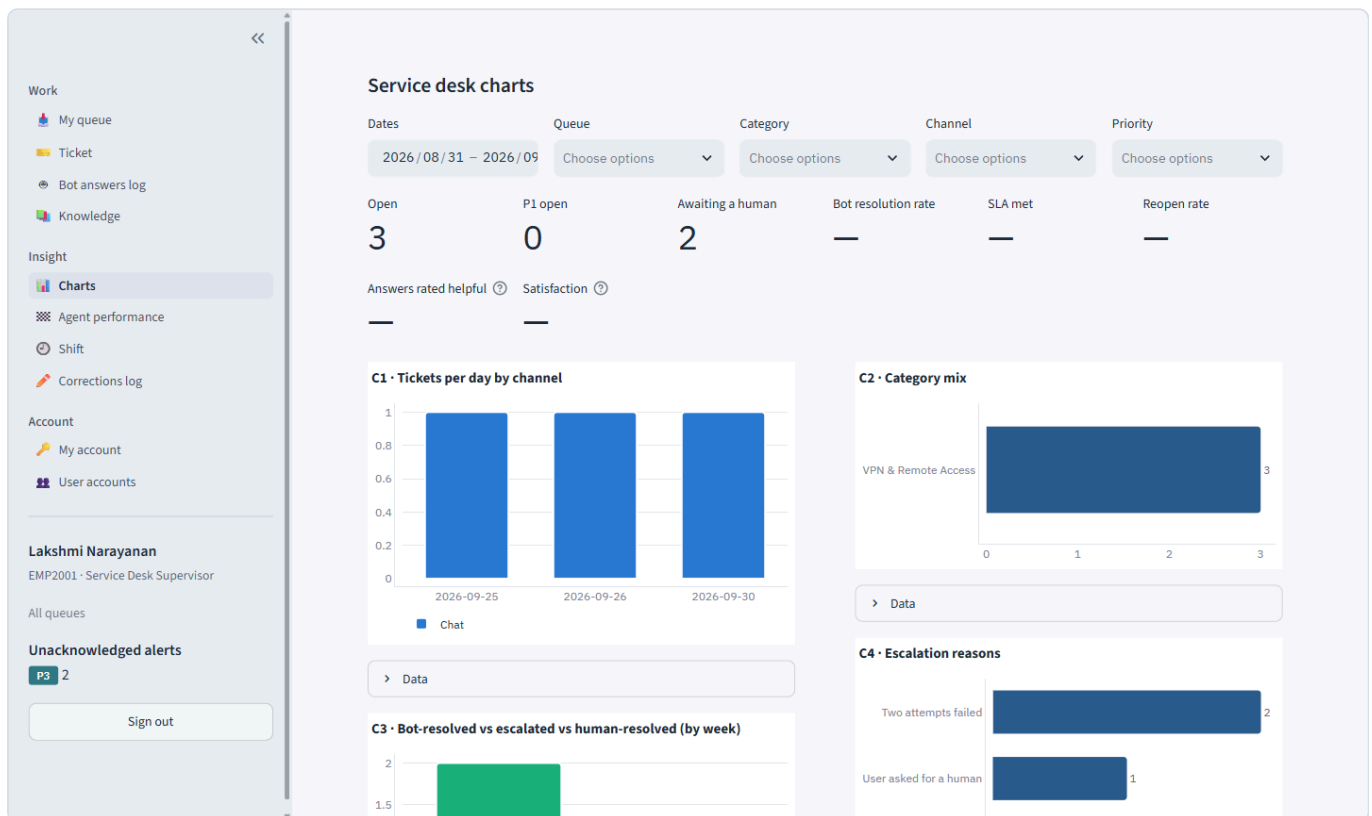


Further down the ticket: bot panel, suggested articles, checklist, and the screenshot with its text.

7. Using it: supervisor

Sign in as **EMP2001**. The supervisor sees every queue, and also:

Page	What for
Charts	Open tickets, P1s, bot resolution rate, SLA met, satisfaction and answers rated helpful, plus the C1–C12 charts (C12 is the safety scorecard)
Knowledge	Approve employee-friendly article drafts (draft → check → approve), per-article helpful %, and the Knowledge gaps list of questions the bot couldn't answer
Agent performance, Shift	Who handled what; the shift handover checklist
Corrections log	Every time an agent fixed the bot's category
User accounts	Create, import, reset, unlock and disable accounts; set agents' queues



Charts, with filters in one row across the top.

<<

Work

My queue

Ticket

Bot answers log

Knowledge

Insight

Charts

Agent performance

Shift

Corrections log

Account

My account

User accounts

Lakshmi Narayanan

EMP2001 - Service Desk Supervisor

All queues

Unacknowledged alerts

P3 2

Sign out

Knowledge

ArticlesKnowledge gapsSaved replies

31 of 31 articles have an approved employee version · 0 draft(s) waiting for approval. Employees see only approved text.

Article	Title	Category	Live	Draft waiting	Helpful %	Ratings
KB-001	Password Reset - Self Service Portal	Password & Account Access	v1			0
KB-002	Assisted Password Reset	Password & Account Access	v1			0
KB-003	Account Unlock - Failed Login Lockout	Password & Account Access	v1			0
KB-004	Multiple Failed Resets - Possible Compromise	Password & Account Access	v1			0
KB-005	VPN Connection Failure - Standard Troubleshooting	VPN & Remote Access	v1			0
KB-006	VPN Certificate / Trust Error	VPN & Remote Access	v1			0
KB-007	VPN Frequent Dropouts	VPN & Remote Access	v1			0
KB-008	New VPN Setup / First Time Install	VPN & Remote Access	v1			0

Open an article

Choose an article

Knowledge: articles, approvals and gaps.

8. Testing with real problems

The automated tests prove that the logic works. Real people phrase things differently, so before the pilot, try these as an employee and note what happens. Use your own words, typos and screenshots from real error messages.

#	Try saying	Should happen
1	"vpn keeps disconnecting every 10 min"	VPN category, OS question, steps, then Network Remote Access
2	A screenshot of a real VPN or Outlook error, with no text	The bot quotes the error line (not the window title) and doesn't ask for the error again
3	"forgot my password and locked out"	Identity flow; it never asks for the password itself
4	"my password is Summer2026! why doesn't it work"	The password is hidden before it's stored, with a warning to change it
5	"outlook not syncing since morning"	Email category with safe steps
6	"printer on 3rd floor shows offline"	Printer category; asks which printer
7	"i clicked a link in an email and entered my login"	Security incident: P1, straight to a person, no self-help steps
8	"laptop smells of burning / battery swelling"	Safety: stop using it, straight to a person
9	"whole office wifi down"	Checks for an existing outage; offers to join the existing ticket
10	"give me admin rights on my laptop, it's urgent"	Never granted by the bot: an approval request goes to a person
11	A message in Tamil or Hindi	Offers a person (the bot answers in English only)
12	Something odd: "how do I fix the coffee machine"	No guessing: a human, plus an entry in Knowledge gaps
13	Start a chat, then press "Talk to a human" immediately	Escalates at once; the agent sees the whole chat
14	As the agent: reply, resolve, then reopen as the employee	Reply shows in the app; reopening goes back to the same queue

For each problem write down: what you typed, the category the bot chose, whether the steps made sense, and whether it escalated correctly. Anything wrong: log a correction as an agent, and add the phrase to `evaluate.py` so it's tested from then on. Run `python evaluate.py --runs 2` after any change.

9. Stop, restart, reset

To	Do
Stop	Press Ctrl+C in each window (tunnel, web app, API)
Restart after changing code	Stop and start the API window again. The web app picks up page changes by itself
Run the tests	<code>.\.venv\Scripts\python.exe -m pytest -q</code> (about 1 minute, all must pass)
Start from an empty database	Stop the API, rename <code>servicedesk.db</code> (keep it as a backup), then start the API again. The demo accounts are recreated
Production restart	<code>dc restart api worker ui</code> ; updates: <code>git pull</code> then <code>dc up -d --build</code>

10. Troubleshooting

You see	Why, and the fix
Blank white page	The web app crashed on load. Check its window for an error. (One cause was fixed on 30 Sep: a file named <code>watchdog.py</code> clashed with a library; it's now <code>healthwatch.py</code> .)
"Can't reach the service desk" / connection errors	The API isn't running. Start window 1 first
Port 8000/8501 already in use	An old copy is still running. Close it, or find it with <code>Get-NetTCPConnection - LocalPort 8501</code> and stop that process
Account locked	5 wrong passwords lock it for 15 minutes. A supervisor can unlock it in User accounts
Bot routes everything to people	The TypeSafe key is missing or wrong, or the service is down (supervisors see a "Model degraded" banner). Check <code>.env</code>
Screenshot "couldn't read"	Photos of a screen are hard to read. A real screenshot (Win+Shift+S) works far better
First screenshot is slow (a few seconds)	The OCR loads on first use. Later ones take 1–4 s
Tunnel link stopped working	The PC slept or the tunnel window closed. Start the tunnel again and share the new link